

T0 Theory: Document Series Overview A Geometric Reformulation of Physics

Systematic Presentation of All 8 Core Documents

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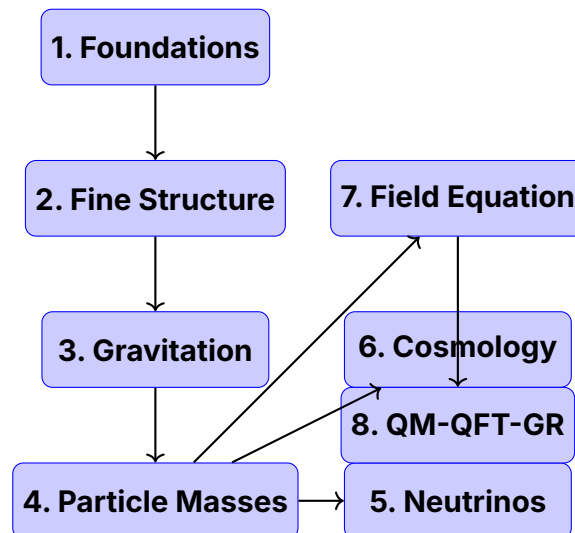
Abstract

This overview presents the complete T0 Theory series consisting of 8 fundamental documents, which represent a geometric reformulation of physics. Based on a single parameter $\xi = \frac{4}{3} \times 10^{-4}$, together with fixed rational structural choices, fundamental constants, particle masses, and physical phenomena from quantum mechanics to cosmology are described within one framework; SI anchors such as E_0 , m_e or ν serve for conversion, not for derivation. Many values are matched to about 1%; because the ladder parameters r_i , p_i are set and one anchor provides the connection to SI, these agreements are not independent evidence. The series deliberately leaves the cosmic sector (H_0 , Λ , dark energy) aside, because the Standard Model does not derive it either; the cosmological statements are speculative. The theory offers testable predictions for future experiments.

.1 Document Series: Systematic Structure

Hierarchical Structure of the 8 Documents

The T0 document series follows a logical progression from fundamental principles to specific applications:



.2 Document 1: 003_T0_Grundlagen_v1_En.pdf

Subtitle: The Geometric Foundations of Physics

Core Contents:

- **Fundamental Parameter:** $\xi = \frac{4}{3} \times 10^{-4}$ as a geometric constant
- **Time-Mass Duality:** $T \cdot m = 1$ in natural units
- **Fractal Spacetime Structure:** $D_f^{\text{eff}} \approx 2.973$ and $K_{\text{frak}} \approx 0.987$
- **Interpretation Levels:** Harmonic, geometric, field-theoretic
- **Universal Formula Structure:** Template for all T0 relations

Fundamental Insights:

- Tetrahedral packing as the fundamental spatial structure
- Quantum field-theoretic derivation of 10^{-4}
- Characteristic energy scales: $E_0 = 7.398 \text{ MeV}$
- Philosophical implications of geometric physics

Status: Theoretical foundation - fully established

.3 Document 2: 011_T0_Feinstruktur_En.pdf

Subtitle: Derivation of α from geometric principles

Central Formula:

$$\alpha = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 \quad (1)$$

Key Results:

- **T0 Value:** $\alpha^{-1} = 137.04$
- **Experiment:** $\alpha^{-1} = 137.036$
- **Deviation:** 0.003%

- **Assessment:** The agreement is a property of the anchor E_0 , which provides the only connection to the SI; it is not an independent prediction.

Theoretical Innovations:

- Characteristic energy $E_0 = \sqrt{m_e \cdot m_\mu / K_{\text{frak}}}$ (bare $\sqrt{m_e m_\mu}$, R72)
- Logarithmic symmetry of lepton masses
- Fundamental dependence $\alpha = \xi E_0^2$ with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}}$
- Why numerical ratios must not be canceled

Status: $\alpha = \xi E_0^2$ as a relation via the anchor E_0

.4 Document 3: 012_T0_Gravitationskonstante_En.pdf

Subtitle: Systematic derivation of G from geometric principles

Complete Formula:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (2)$$

Conversion Factors:

- **Dimensional Correction:** $C_1 = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ with $E_{\text{char}} = 28.8$ (Doc. 012); C_1 does not appear in the boxed formula above, which holds without C_1
- **SI Conversion:** $C_{\text{conv}} = 7.783 \times 10^{-3}$
- **Fractal Correction:** $K_{\text{frak}} = 0.986$

Experimental Verification:

- **T0 Prediction:** $G = 6.6745 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$
- **CODATA 2018:** $G = 6.67430 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$
- **Deviation:** +0.003%

The boxed formula with the stated factors gives 6.6745×10^{-11} . C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383); with the anchor m_e , G lies at -2.6% , with v at -0.46% . The agreement to 0.003% arises only with this residual and does not count as an independent precision result.

Physical Meaning: Gravity as geometric spacetime-matter coupling

Status: Form and order of magnitude from ξ and one anchor; the CODATA agreement is not an independent precision result

.5 Document 4: 006_T0_Teilchenmassen_En.pdf

Subtitle: Fermion masses from the ladder $m_i = r_i \xi^{p_i} v$

Two Equivalent Methods:

1. **Direct Method:** $m_i = 1/(\xi f_i)$, linked to the Yukawa method via $f_i = 1/(r_i \xi^{p_i+1} v)$; the agreement of both methods holds by construction
2. **Extended Yukawa:** $m_i = y_i \times v$ with $y_i = r_i \times \xi^{p_i}$

Quantum Number System: Rational quantum numbers (r_i, p_i) as fixed structural choices; v serves as conversion anchor

Comparison with Experiment: For charged leptons and quarks the average deviation is below 1%. Because r_i and p_i are set and adjusted to the measured values, these agreements are identities of the ladder and not evidence in their own right; the mass ratios follow from the ladder to a few digits. Neutrino masses are not calculated in Doc. 006 (Doc. 340 is authoritative); boson masses are treated in Doc. 372 and 373.

Reduction: The free mass parameters of the Standard Model are traced back to ξ and the set rational quantum numbers (r_i, p_i) ; no continuously adjustable parameter remains.

Status: Ladder structure with set quantum numbers and one anchor

.6 Document 5: 007_T0_Neutrinos_En.pdf

Subtitle: The Photon Analogy and Geometric Oscillations

Approach:

- **Photon Analogy:** Neutrinos as 'damped photons'
- **Double ξ -Suppression:** base value $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54 \text{ meV}$
- **Geometric Oscillations:** Phases instead of mass differences

Experimental Context:

- **Spectrum:** A spectrum that is (nearly) uniform for all flavors, with $\Sigma m_\nu \approx 13.6 \text{ meV}$, is excluded, because the measured mass differences require $\Sigma m_\nu \geq 59 \text{ meV}$.
- **Velocity:** An energy-independent difference $v_\nu = c(1 - \xi^2/2)$ does not hold, because $\xi^2/2$ is the mass ratio m_ν/m_e .
- **Authoritative:** For neutrinos, Doc. 340 is authoritative; there the base value 4.54 meV reappears as the lightest mass m_{ν_1} .

Status: Speculative; the spectrum of Doc. 007 is excluded by the oscillation data, Doc. 340 is authoritative

.7 Document 6: 025_T0_Kosmologie_En.pdf

Subtitle: Static Universe and ξ -Field Manifestations

Cosmology:

- **Static Universe:** No Big Bang, eternally existing
- **Time-Energy Duality:** No singular beginning, because $T \cdot E = 1$ with the lower time bound $T_0 = \xi t_P$ bounds the energy from above and the torus has no boundary (Doc. 306)
- **CMB from ξ -Field:** Not from $z=1100$ decoupling

Casimir-CMB Connection:

- **Characteristic Length:** $L_\xi = 100 \text{ } \mu\text{m}$
- **Ratio at L_ξ :** $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = 308$ – an identity, not a measurement result, because L_ξ itself is determined from the measured CMB energy density and ξ ; a comparison value of 312 is the same formula with a rounded L_ξ . The connection is a statement about scales (speculative).
- **In depth:** Doc. 279 – Casimir-CMB- H_0 connection formula, $L_\xi H_0/c = K_{\text{geo}}(m_e c^2/k_B T_{\text{CMB}})\xi^{41/4}$.

Alternative Redshift (static, achromatic):

$$1 + z = e^{\xi x}, \quad z \approx \xi \cdot x \quad (3)$$

The fractal path lengthening is purely geometric and stretches all wavelengths by the same factor; the redshift is *achromatic* (finite-element simulation). The length scale of x does not follow from ξ alone but is calibrated via the measured H_0 . In depth: Doc. 280 (z→time as a Λ CDM translation, cosmological degeneracy Doc. 267, $z_* \approx 875$ as a T^4 lattice resonance).

Cosmological Problems (speculative):

- Horizon problem, flatness problem, monopole problem, age problem
- H_0 and Λ do not follow from ξ alone: the scale of the redshift is calibrated via the measured H_0 . Hubble tension and dark energy are therefore not regarded as solved; the cosmic sector is deliberately left aside, because the Standard Model does not derive it either.

Status: Testable hypotheses - speculative alternative, cosmic sector calibrated

.8 Document 7: 019_T0_lagrndian_En.pdf

Subtitle: The Fundamental Field Equation of T0 Theory

Core Contents:

- **Fundamental Field Equation:** $\square\Phi + \xi \cdot \mathcal{F}[\Phi] = 0$
- **Geometric Interpretation:** Field as manifestation of spatial structure
- **Unification:** All interactions from one equation
- **Solution Classes:** Particle solutions, wave solutions, vacuum solutions

Physical Consequences:

- Emergent gauge symmetries from geometric constraints
- Quantization as a natural consequence of field geometry
- Scale structure as flow on the recursion $\mathcal{R}$ in parameter space
- Phase transitions as topological changes

Status: Theoretical framework - fully formulated

.9 Document 8: 020_T0_QM-QFT-RT_En.pdf

Subtitle: Unification of QM, QFT, and GR from a geometric foundation

Core Contents:

- **Universal T0 Field Equation:** $\square\Phi + \xi \cdot \mathcal{F}[\Phi] = 0$ as the basis of all theories
- **Time-Mass Duality:** $T \cdot m = 1$ connects all three pillars of physics
- **Emergent Quantum Properties:** QM as approximation of the energy field
- **Field Description:** All particles as excitations of a fundamental field Φ
- **Renormalization Solution:** Natural cutoff by E_P/ξ

Fundamental Insights:

- Deterministic interpretation of quantum mechanics via the time field (speculative; local only under the assumption of superdeterminism, the Bell violation remains)

- Wave-particle duality from field geometry
 - Energy scale hierarchy: Planck down to QCD via ξ -corrections
 - Gravity as field curvature
 - Philosophical implications: Unity of physics through geometric principles
- Status:** Theoretical unification - builds on all previous documents, testable predictions

.10 Comparison and positioning series (IPI field)

Complementing the foundational series above, a separate series documents the position of FFGFT relative to other frameworks discussed at the Information Physics Institute. These documents are *comparisons and placements*, not new foundations:

- **Doc. 245** – FFGFT mapped onto RA 2.1 (Guevara).
- **Doc. 246** – FFGFT vs. RSG (Austin): asymmetric complementarity.
- **Doc. 251** – FFGFT vs. Vopson (MEI / Computational Universe): geometry- vs. information-primitive.
- **Doc. 260** – FFGFT vs. UC (Haskins): constraint, α -correction inconsistency.
- **Doc. 269** – FFGFT $\leftrightarrow$ RA/PMT (Guevara), bidirectional.
- **Doc. 271** – FFGFT $\leftrightarrow$ HLV (Krüger), pairwise comparison; the spectral fork.
- **Doc. 272** – FFGFT $\leftrightarrow$ HLV via the Dot governance layer (Vossen).
- **Doc. 274** – FFGFT in the IPI field: substrate and primitive axes (map); A* worked example.

Status: ongoing correspondence series, dated and provisional; each placement is a declared reading (O2), the primary rendering (O0) remaining with each originator.